Traffic Engineering

METI

Lab 1 Report [EN]

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Group 5

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1 Goals

The objective of this laboratory assignment was to simulate and analyse the behaviour of Poisson arrival processes and compare the experimental results with their theoretical probability distributions.

The work developed so far includes:

- Simulation of independent Poisson arrival processes;
- Manual construction of histograms without using built-in histogram functions;
- Comparison between experimental and theoretical Poisson distributions;
- Analysis for different values of λ;
- Superposition of multiple independent Poisson processes.

The implementation was developed in Python using random number generation, numerical functions and graphical visualization tools.

2 Poisson Process Simulation

2.1 Theoretical Background

A Poisson process with parameter λ represents a random arrival process in which the number of events occurring in a unit time interval follows a Poisson distribution:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \tag{1}$$

where:

- k = number of events in one interval;
- λ = average arrival rate.

The time between two consecutive arrivals follows an exponential distribution:

$$f(t) = \lambda e^{-\lambda t}, \quad t \geq 0 \tag{2}$$

To generate exponentially distributed inter-arrival times, the inverse transform method was used:

$$\Delta t = \frac{-\ln(1-u)}{\lambda} \tag{3}$$

where u is uniformly distributed in $[0, 1)$.

2.2 Implementation Method

For each experiment:

1. Generate N inter-arrival times Δt;
2. Compute cumulative arrival instants;
3. Divide time into unit intervals;
4. Count the number of arrivals in each interval;
5. Build the histogram manually;
6. Compare experimental probabilities with the theoretical Poisson law.

The use of built-in histogram functions was intentionally avoided, as requested in the assignment statement.

2.3 Results for Different Values of λ

The simulator was executed for:

$$\lambda = \{0.5,\ 1,\ 5,\ 10,\ 50\}$$

For each value, the number of generated samples was increased proportionally in order to reduce statistical noise for large λ values.

$\lambda = 0.5$

For small values of λ, most intervals contain zero or one arrival. This behaviour matches the theoretical expectation, where $P(0)$ is dominant.

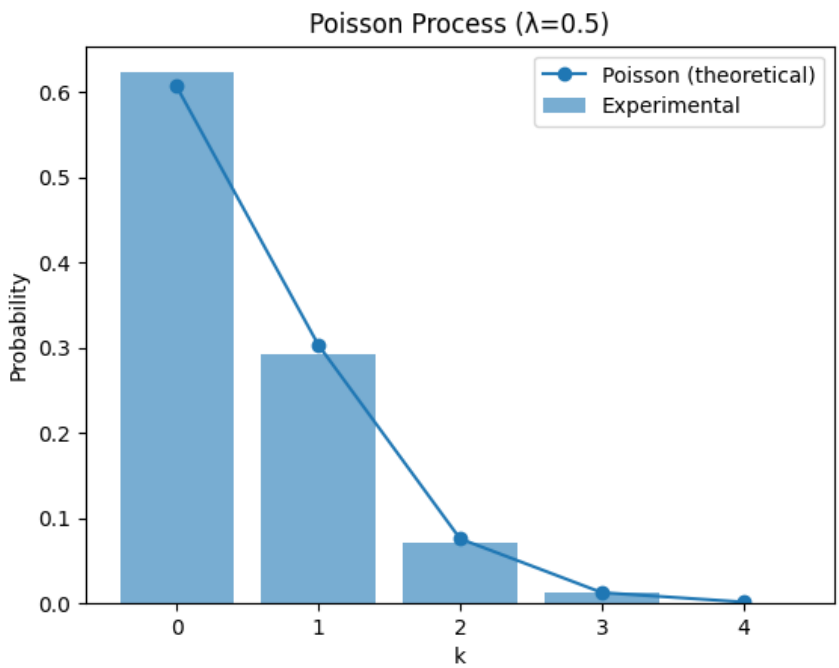

Figure 1: Experimental and theoretical distribution for $\lambda = 0.5$.

The probability mass remains concentrated at low values of k, although intervals with two or more arrivals become more frequent.

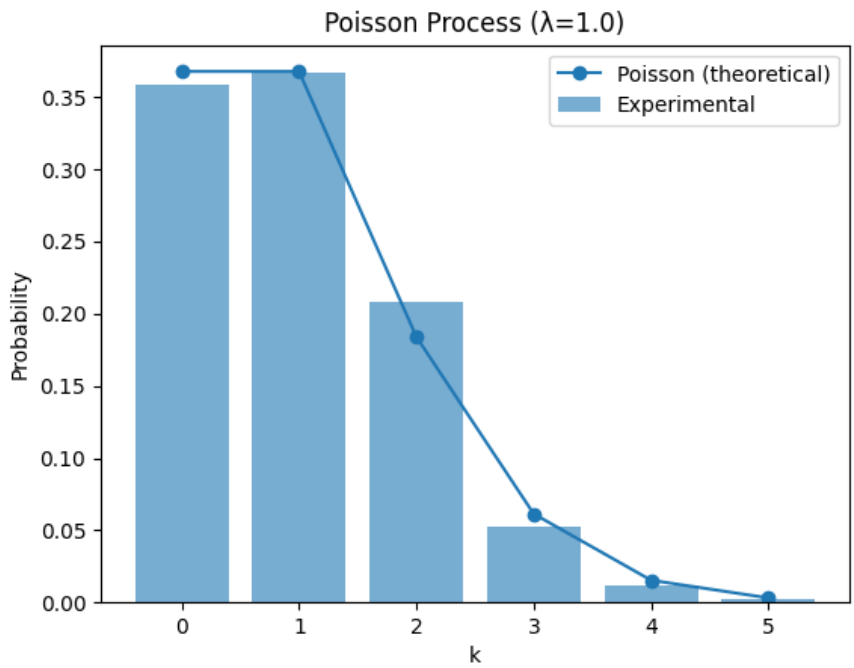

Figure 2: Experimental and theoretical distribution for $\lambda = 1$.

The distribution becomes wider and more symmetric around the mean value.

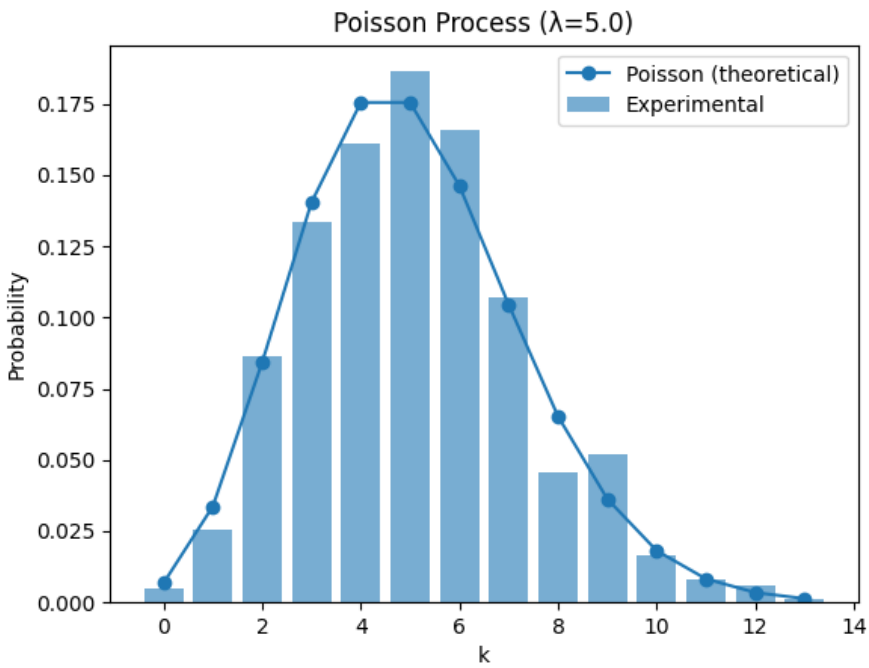

Figure 3: Experimental and theoretical distribution for $\lambda = 5$.

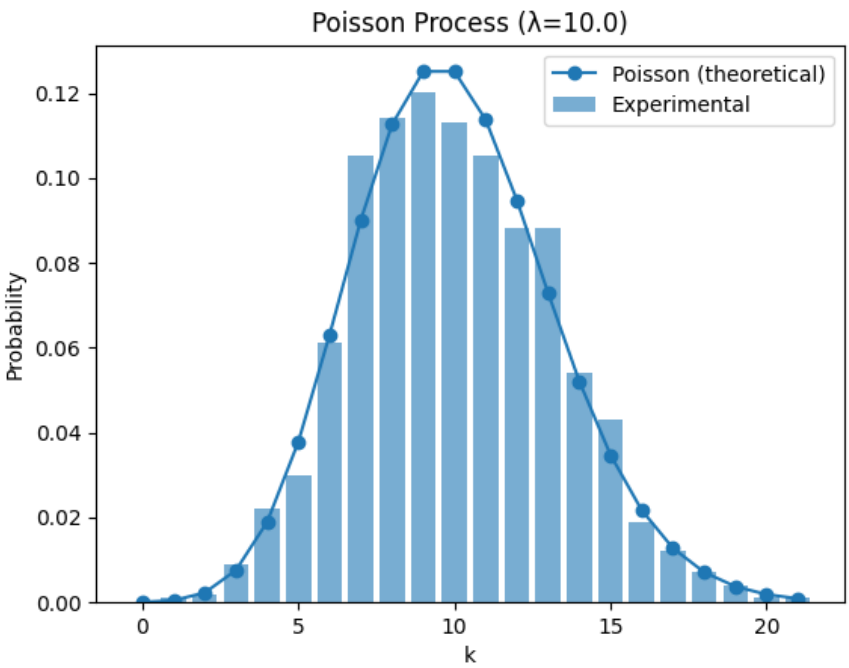

Figure 4: Experimental and theoretical distribution for $\lambda = 10$.

For very large arrival rates, the Poisson distribution becomes approximately bell-shaped due to convergence towards a normal distribution.

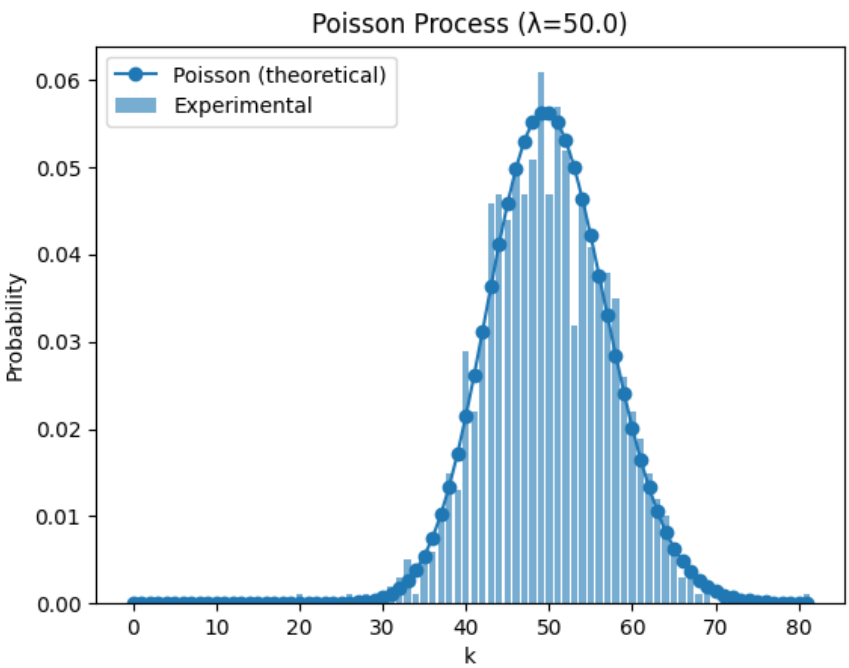

Figure 5: Experimental and theoretical distribution for $\lambda = 50$.

2.4 Discussion

The obtained results validate the simulator implementation. As the number of generated events increases, the experimental distribution converges to the theoretical model.

It was also observed that:

- Small λ values produce sparse arrival patterns;
- Larger λ values generate more regular and concentrated distributions;
- Increasing the sample size improves accuracy and reduces random fluctuations.

3 Superposition of Poisson Processes

3.1 Theoretical Background

A fundamental property of Poisson processes states that the superposition of independent Poisson processes is also a Poisson process whose rate is equal to the sum of the individual rates.

Given four independent processes:

$$\lambda_1 = 2, \quad \lambda_2 = 9, \quad \lambda_3 = 11, \quad \lambda_4 = 15$$

the resulting combined process has parameter:

$$\lambda_{total} = 2 + 9 + 11 + 15 = 37 \tag{4}$$

3.2 Implementation

Four independent Poisson processes were generated using exponential inter-arrival times with parameters $\lambda_1, \lambda_2, \lambda_3$, and λ_4.

Instead of fixing a predefined number of events for each process, all processes were simulated over a common observation interval T. This approach ensures that the resulting number of events in each process is consistent with the statistical properties of a Poisson process, where the number of arrivals in a time interval is itself a random variable.

The choice of a fixed observation interval is essential for correctly demonstrating the superposition property. In a Poisson process, the expected number of arrivals in a time interval T is given by:

$$\mathbb{E}[N(T)] = \lambda T$$

Thus, for each individual process, the number of generated events naturally depends on both the rate λ and the observation interval T. Processes with higher λ values produce, on average, more events within the same interval.

After generating the four independent arrival sequences over the same interval T, their arrival instants were merged into a single sequence and sorted in chronological order. This produces a valid combined arrival process.

A histogram of the number of events per unit time interval was then constructed using the same methodology described in Section 2.2. This histogram was compared against the theoretical Poisson distribution with parameter:

$$\lambda_{total} = \sum_i \lambda_i$$

3.3 Results

In practice, a fixed number of events N was generated for each process, which implicitly defines a random observation interval T. For sufficiently large N, this approach provides a good approximation of a Poisson process observed over a fixed time window.

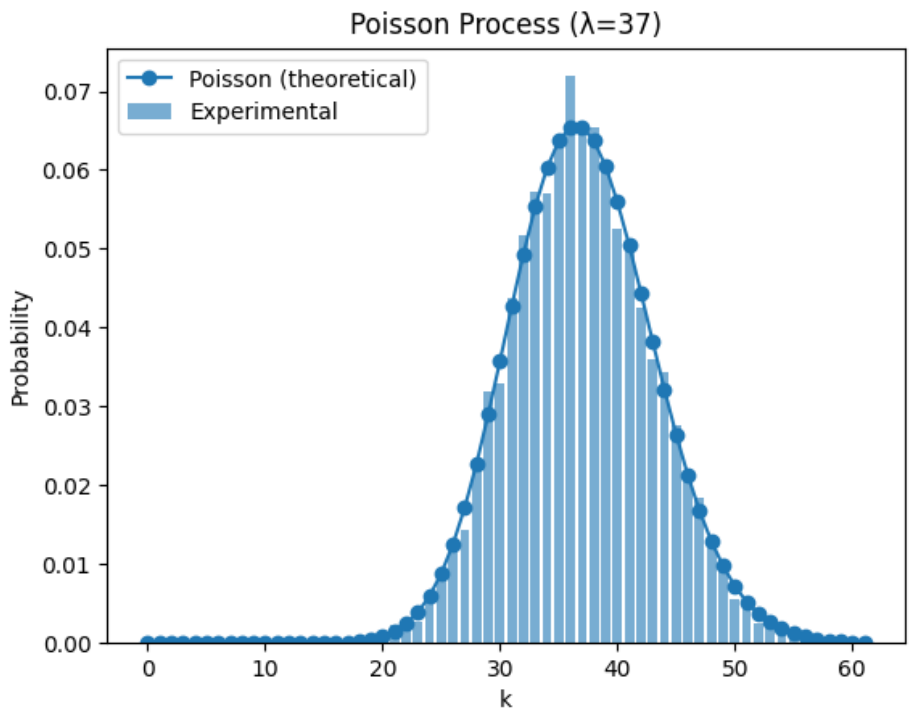

Figure 6: Experimental and theoretical distribution for the superposed process ($\lambda = 37$).

The experimental histogram is centered near the expected mean value and follows the theoretical Poisson distribution with good accuracy.

3.4 Discussion

This experiment confirms the closure property of Poisson processes under superposition. The sum of independent arrival streams preserves the Poisson nature of the process while increasing the global arrival rate.

Such behaviour is highly relevant in communication networks, where traffic generated by multiple independent users can often be modelled as a single aggregate Poisson flow.

4 Conclusion

The objectives defined up to Section 2.3 of the assignment were successfully completed. The developed simulator correctly reproduces:

- Individual Poisson arrival processes;
- Theoretical event-count distributions;
- Statistical convergence with larger sample sizes;
- Superposition of independent Poisson processes.

The methodology developed in this work can be extended to more complex traffic models, enabling further exploration of stochastic processes in real-world systems.

5 M/M/1 Queue Simulation

5.1 Theoretical Background

An M/M/1 queue models a system with:

- Poisson arrivals with rate λ;
- Exponentially distributed service times with rate μ;
- A single server;
- FIFO (First-In-First-Out) discipline.

A key parameter of the system is the utilization factor:

$$\rho = \frac{\lambda}{\mu} \tag{5}$$

The behaviour of the system depends on the value of ρ:

- $\rho < 1$: stable system;
- $\rho = 1$: critical regime;
- $\rho > 1$: unstable system.

For a stable system ($\rho < 1$), the main theoretical results are:

$$L_q = \frac{\rho^2}{1-\rho} \tag{6}$$

$$W_q = \frac{\rho}{\mu - \lambda} \tag{7}$$

$$W = \frac{1}{\mu - \lambda} \tag{8}$$

$$U = \rho \tag{9}$$

where:

- L_q = average queue length;
- W_q = average waiting time in queue;
- W = average time in system;
- U = server utilization.